



SCIENCE

REVIEW

March 3-4, 2026

User / Scientific Community Outreach

Arpita Patel

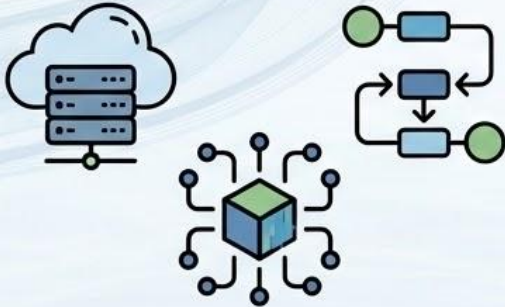
Alabama Water Institute, CIROH

The University of Alabama

Research Theme 2 (RT2): Community Water Modeling

Enabling collaborative development, testing, and deployment of hydrologic models through shared frameworks and infrastructure — central to CIROH's support of the NextGen Water Resources Modeling Framework.

Community Cyberinfrastructure



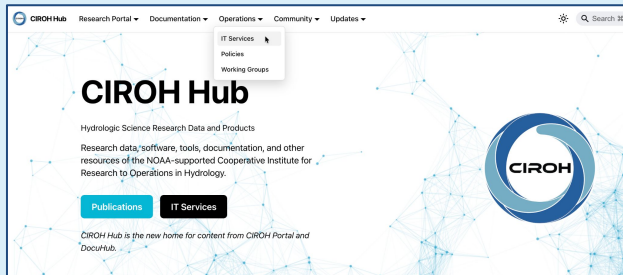
Cloud platforms, HPC, workflow automation, and shared data services — built for accessibility, scalability, and collaborative research.

Community Modeling



Open-source model development, integration, and governance through containerization, shared codebases, and CI/CD practices.

CIROH Hub Blogs for Community Water Modeler



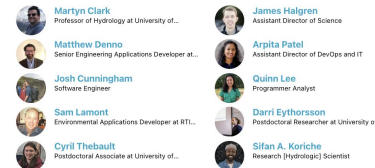
AWS re:Invent 2025: Key Insights for Research and Cyberinfrastructure

January 2, 2026 · 4 min read



Moving Hydrologic Prediction Forward — A software integration meeting at the Alabama Water Institute

November 24, 2025 · 10 min read



Google Cloud Next 2025: Innovation at Scale

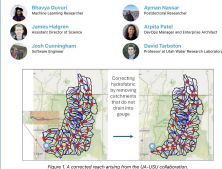
April 16, 2025 · 5 min read



If you missed the keynote, I highly recommend watching the recording here: [GCN25 Keynote Video](#)

Cross-Institutional Collaboration Enhances Hydrologic Modeling in the Logan River Watershed

August 18, 2025 · 8 min read



Hourly Differentiable Modeling Arrives in the NGIAB-NRDS NextGen Ecosystem

February 20, 2026 · 9 min read



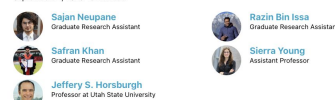
AORC Data in Your Hands: User-Friendly Jupyter Notebooks for Data Retrieval and Analysis via CIROH-2i2c JupyterHub Notebooks

July 1, 2025 · 3 min read



Focusing on Streamflow Data: New Software for Camera-Based Hydrologic Modeling

September 24, 2025 · 5 min read



Evaluating NextGen's Performance in the MARFC Region with NGIAB

August 28, 2025 · 7 min read



The National Weather Service's Middle Atlantic River Forecast Center (MARFC) sees large variations in the performance of the National Water Model (NWM). Through its support for regionalized parameters and models, NOAA-Cyberinfrastructure Water Resources Modeling Framework (DevOps Framework) offers a potential solution to address these inconsistencies. As such, this study tests an integration of hardware in the NWM to evaluate the DevOps Framework's performance in the MARFC region.

This study evaluated three operational hydrologic modeling frameworks (applied at the National Water Model (NWM) the Community Hydrologic Prediction System (CHIPS), the NextGen framework, and version 3.0 of the National Water Model (NWM)).

- CHIPS is the current operational framework used by NOAA's River Forecast Centers. It incorporates the SWAT-IT model for watershed and the Sacramento Soil Moisture Accounting (SSA) model for runoff generation.
- For the early phases of this study, the NextGen framework was used with the default model configuration provided by the NWM, which includes the Hydro-Cyberinfrastructure (H-Cyber) and surface model and the Computational Functional Equations (CFE) (variational runoff) model [1].
- After initial runs with the baseline configuration, Next-Gen-Modeler was reconfigured with SWAT-IT output and simplified Potential Evapotranspiration (PET) values from the MARFC database.

Operations Community Updates

CIROH Hub Blog

Exclusive content for researchers utilizing CIROH Cyberinfrastructure resources. Share your insights, discoveries, and experiences with the hydrologic science community. This blog platform is dedicated to highlighting the innovative work of researchers who have leveraged CIROH computational tools and resources to advance water science. You share the knowledge and expertise of our CIROH researchers and inspire new applications across the field.

Submit Your Blog

2026

Hourly Differentiable Modeling Arrives in the NGIAB-NRDS NextGen Ecosystem CIROH Hub, New York Central Home for CIROH Research, Tools, and Community Resources

CIROH Cyberinfrastructure and Hydroinformatics Team at ARCC

ARCC is a 2025. We brought the Research and Cyberinfrastructure

2025

Moving Hydrologic Prediction Forward — A software integration meeting at the Alabama Water Institute

Building Bridges: CIROH's New Data Collaboration Framework Differentiates Modeling in the NWM

From Researcher's Impact: CIROH Science Hardware Data Resources and Collections

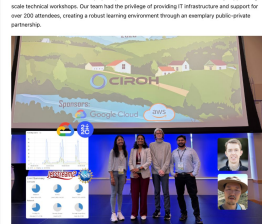
DevCon 2025: A DevOps and Cyberinfrastructure Success Story

June 4, 2025 · 3 min read

Arpita Patel

DevOps Manager and Enterprise Architect

The recent DevCon 2025 event showcased not just cutting-edge development practices, but also demonstrated how modern DevOps principles and cloud infrastructure can seamlessly support large-scale technical ecosystems. Our team led the planning of providing IT infrastructure and support for over 200 attendees, creating a robust learning environment through an exemplary public-private partnership.



40+

Blogs so far!
More to come

Pennsylvania State University Researchers Leverage CIROH Cyberinfrastructure for Advanced Hydrological Modeling

February 28, 2025 · 3 min read



Pennsylvania State University (PSU) researchers have been leveraging CIROH Cyberinfrastructure to tackle complex hydrological modeling challenges. This post highlights their innovative approach using the [Wulokun computing platform](#) in conjunction with Amazon S3 bucket storage to efficiently process and analyze large-scale environmental datasets.



<https://hub.ciroh.org/blog>

CIROH User Support

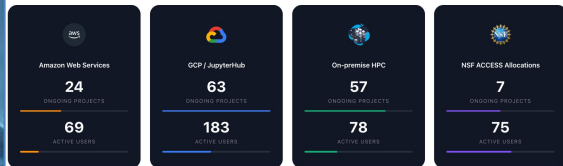
CIROH Hub: Documentation & Infrastructure Access

CIROH Cyberinfrastructure & Community NextGen Monthly Office Hours

Join us for our monthly sessions focused on CIROH's research computing resources and the Community NextGen project. These office hours provide an excellent opportunity to engage with experts, ask questions, and stay updated on the latest developments.

How to Join

Email us at ciroh-r-admin@uia.edu to receive the Teams Meeting link and calendar invitation.



Community NextGen Updates

Stay connected with the latest developments in NextGen water modeling! This news hub brings you updates, breakthroughs, and opportunities from across our community of practice.

YEAR

2026

MONTH

February

Expand all months

Support Channels



Monthly Community NextGen Updates



AWS Office hours



CIROH Slack Channels

Key Benefits



Enables R2O accelerations



Enabling latest technology for Community Water modeling

